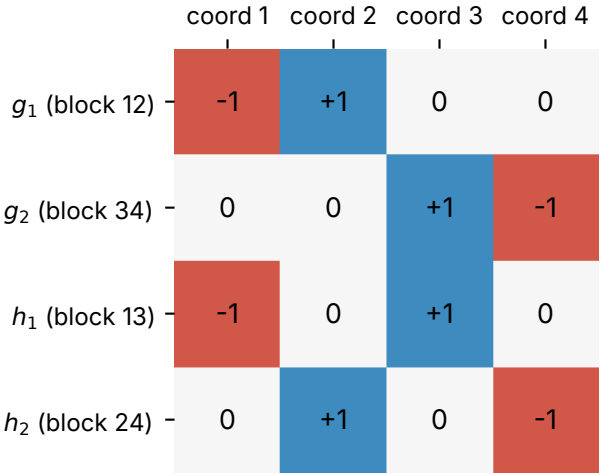

$g_1 + g_2 = h_1 + h_2 = (-1, 1, 1, -1)$, rank = 3 (base point $x = 1001$, exact elimination)